PHY101

Quiz no 5

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What happens to the total capacitance when capacitors are connected in series?

It decreases

It becomes infinite

It remains the same

It increases

How does increasing the intensity of light in the photoelectric effect influence the number of ejected electrons?

Decreases the kinetic energy of the photoelectrons.

Changes the color of the emitted light.

No effect on the number of photoelectrons.

Increases the number of photoelectrons.

The primary purpose of the equation of state in thermodynamics is:

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To define the system's internal energy

To relate thermodynamic variables like P, V, and T

To explain the conservation of energy

To determine the path variables

Which of the following electromagnetic radiations has photons with the greatest momentum?

radio waves

x rays

yellow light

blue light

If the source temperature is raised, the efficiency of heat engine will:

Remain constant

First increase then decrease

Decrease

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Increase

Which of the following types of electromagnetic radiation travels at the greatest speed in vacuum?

X rays

Radio waves

Gamma rays

All of these travel at the same speed

In relativistic mechanics, the rest energy \[E_0\] is equal to:

\[{E_0} = \sqrt {\gamma {m_0}{c^2}}\]

E0} = m_0c^2

\[{E_0} = \frac{1}{2}{m_0}{u^2}\]

\[{E_0} = \sqrt {{m_0}{c^2}}\]

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Fiber optics works on the principle of _____.

Fermat's principle

Total internal reflection

Young's principle

Snell's law

A thermometer indicates 98.6?C.It may be:

Outdoors on ac old day

In a cup of hot tea

In a comfortable room

In a normal person’s mouth

When light is incident perpendicular to the surface, what will be the angle of incidence and the angle of reflection?

Angle of incident = 45°, Angle of reflection = 0°

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Angle of incident = 0°, Angle of reflection = 90°

Angle of incident = 0°, Angle of reflection = 0°

Angle of incident = 90°, Angle of reflection = 0°

It is more di?cult to measure the coe?cient of volume expansion of a liquid than that of a solid because:

The containing evessel also xpands

No relation exists between linear and volume expansion coe?cients

A liquid tends to evaporate

A liquid expands too much when heated

Which celestial body is involved in Ole Rømer's experiment to measure the speed of light?

Mars

Saturn

Venus

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Jupiter

Electromagnetic waves were experimentally discovered by:

Ampere

Maxwells

Hertz

Einstein

Due to relative motion of observer and the frames of reference of events, time always:

Continuous itself

Statics itself

Stretches itself

Contracts itself

The photoelectric effect is described as:

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Electrons are ejected from a metal surface under the influence of light.

Electrons move freely in a metal under the influence of an electric field.

Metals emit light when electrons move from one energy level to another.

Electrons are emitted from a metal surface when it is heated.

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?

Current

Energy

e.m.f

Power

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Which of the following is usually taken to make the core of a transformer?

Hard iron

Soft iron

Aluminum

Copper

What happens to the electric field at the surface of a perfect electric conductor

Electric field is continuous and proportional to sigma

Electric field is discontinuous and proportional to sigma

Electric field is zero

Electric field is infinite

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Force per unit charge is:

Lorentz force

Electric field intensity

Electric Flux

Electric potential

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